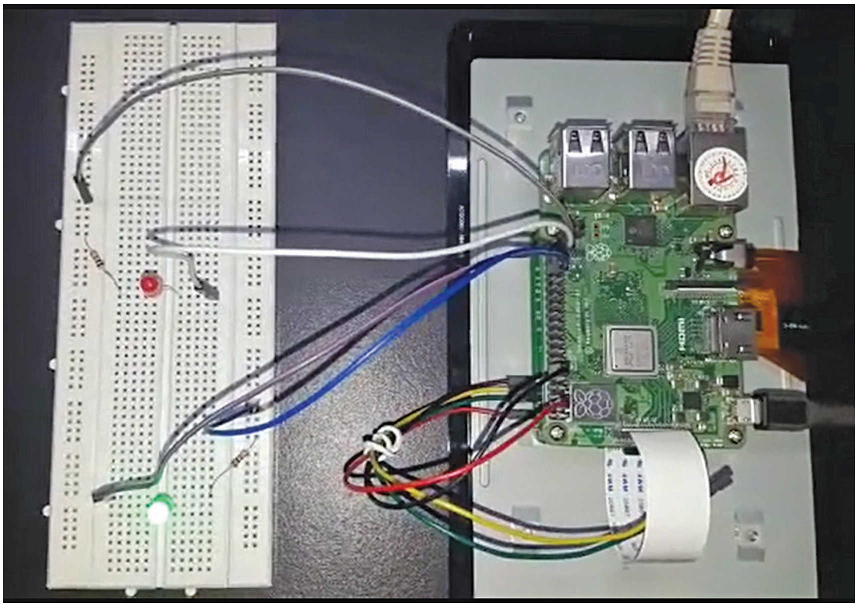

Fig. 5: GPIO pins connected to LEDs on breadboard

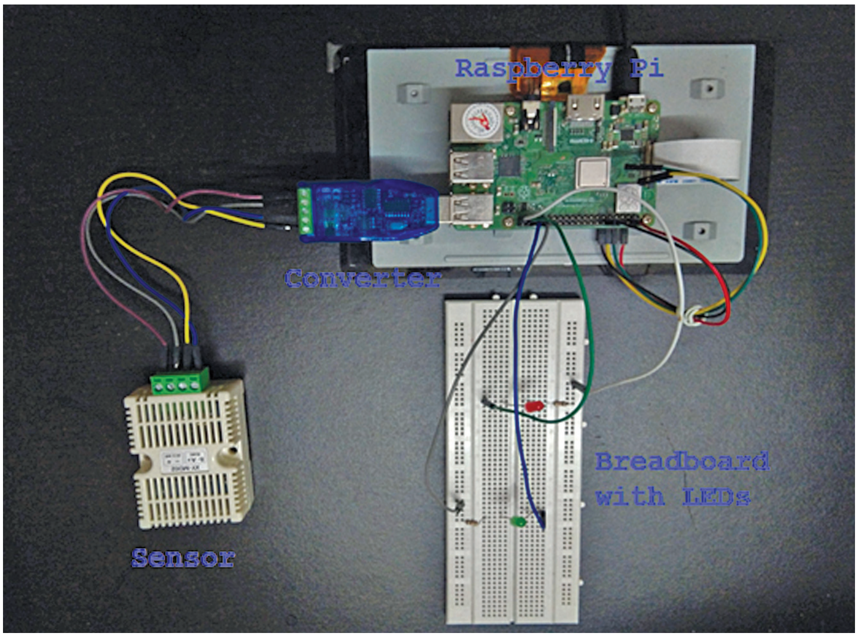

Fig. 6: Physical connections for end-to-end data flow

humidity transmitter SHT20 sensor high precision monitoring Modbus RS485. Sensor SHT20 sends temperature and humidity values with a multiple of 10. After receiving the values, you need to divide them by 10 to get the correct readings. Industrial USB-to-RS485 bidirectional half duplex serial line converter is used for sending and receiving values from the sensor. Jumper wires are used to connect the sensor to converter. Fig. 3 shows connection of SHT20 sensor to Raspberry Pi using USB-to-RS485 converter.

Reading values

Our objective is to check the functionality of the sensor without using EdgeX, so that we can validate whether the sensor is working as expected or not. The author tested the Modbus messages from sensor with a C program. In C program, libmodbus library is used, which provides the APIs to send and receive messages in Modbus protocol.

```
--------------------------------------------
root@ubuntu:/home/ubuntu/test# gcc -o readVaules read_sensor_values.c `pkg-config --cflags --libs libmodbus`
root@ubuntu:/home/ubuntu/test# ./readVaules
Setting slave_id 1
Opening /dev/ttyUSB0 at 9600 bauds (N, 8, 1)
[01][04][00][01][00][02][20][0B]
Waiting for a confirmation...
<01><04><04><01><4A><02><19><1B><04>
Temperature = 33.000000 & Humidity = 53.700000
root@ubuntu:/home/ubuntu/test#
--------------------------------------------
```

Decoding Modbus messages

Let’s decode a pair of messages sent to and received from the sensor. Request sent to the sensor: 01 04 00 01 00 02 20 0B

Sent	Decoding
01	Slave ID
04	Function, read_input_registers
00 01	Register address
00 02	Number of registers to read
20 0B	CRC

Response received from the sensor: 01 04 04 01 4A 02 1A 5B 05

Received	Decoding
01	Slave ID
04	Function, read_input_registers
04	Bytes count
01 4A	Data 1, Temperature
02 1A	Data 2, Humidity
5B 05	CRC

Temperature value received is 0x014a, which is 330 in decimal. SHT20 sends the values in multiple of 10, so temperature is 33.0 degrees.